CODSOFT DATA ANALYTICS INTERNSHIP

TASK 3

Data Visualization Dashboard

Retail Sales Insights using Python and Matplotlib

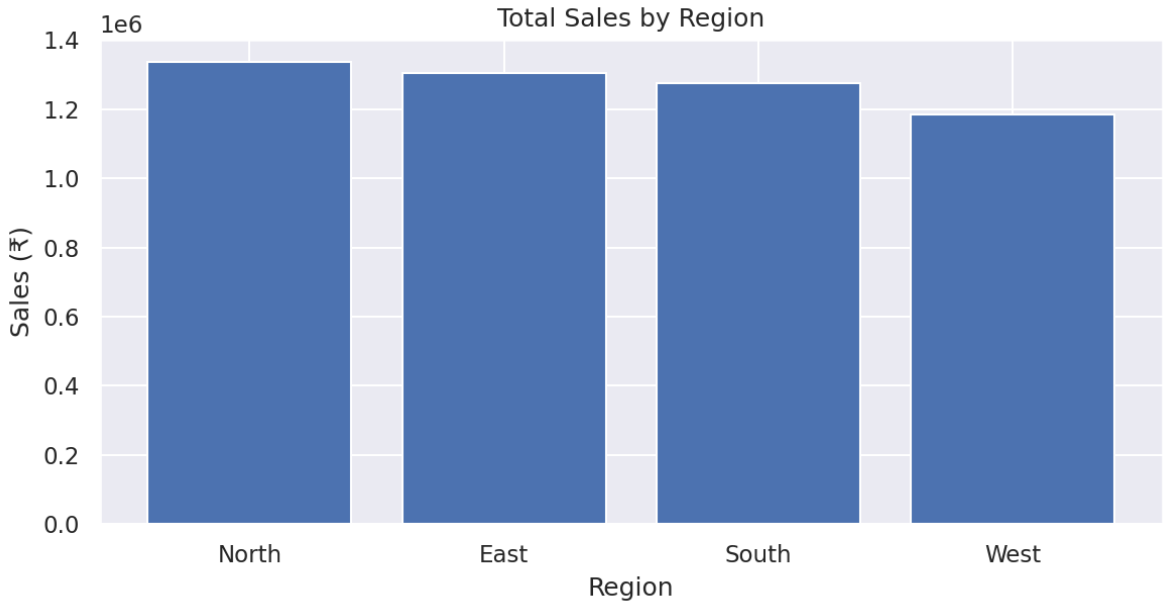
Total Sales Records	Total Sales	Regions	Categories
48	■5,100,000	4	3

Project objective

This project presents sales data in clear charts to make it easier to compare regions, monitor monthly trends, understand category contribution, and inspect the distribution of sales values.

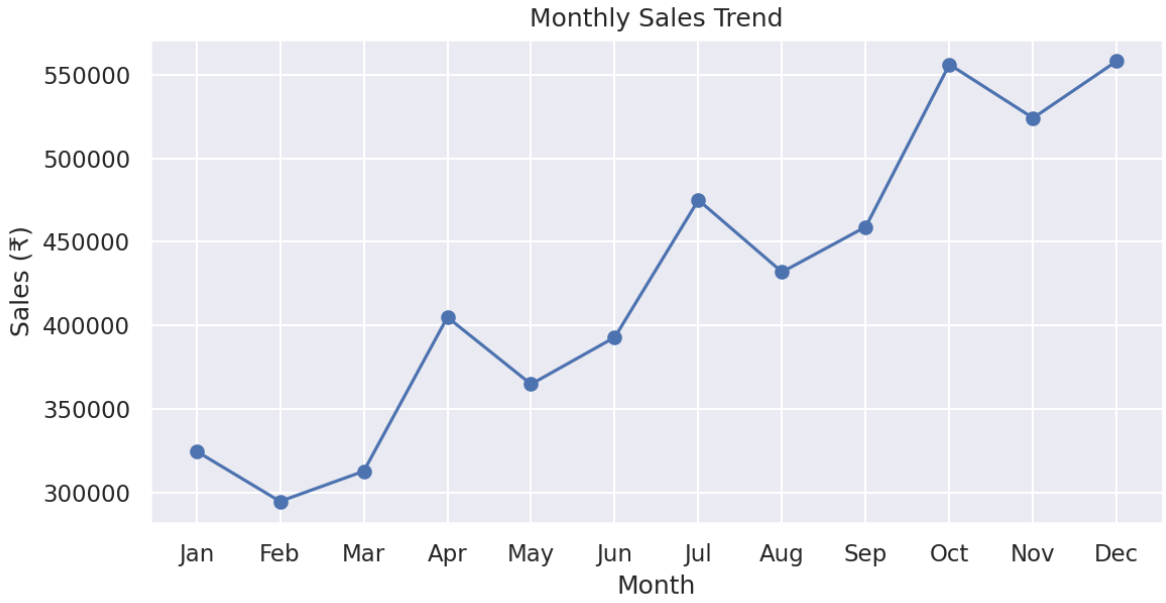
Tools used: Python, Pandas, Matplotlib. The dataset is synthetic practice data created for this learning project.

1. Sales by Region

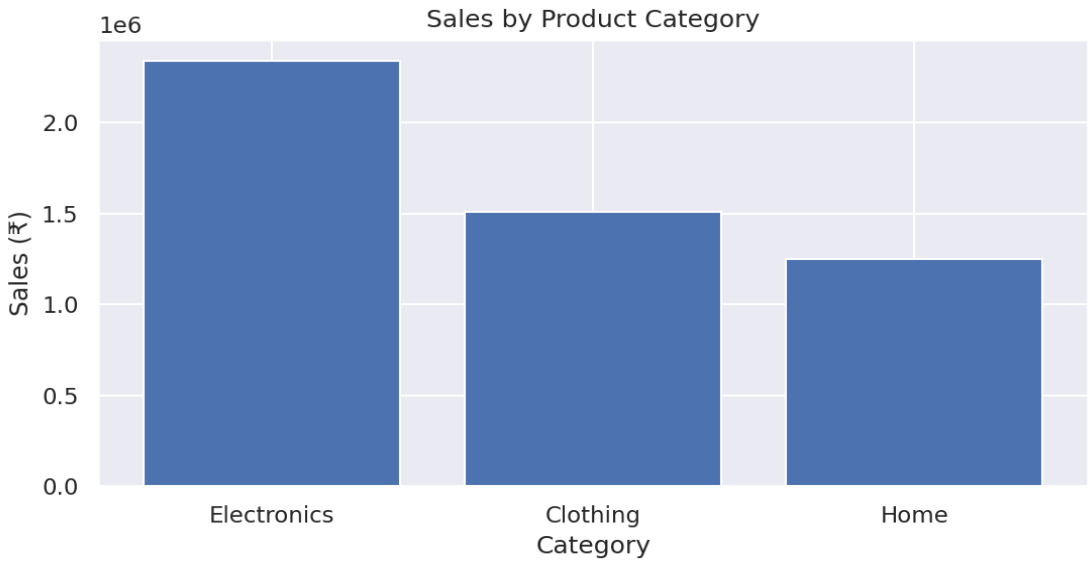


Finding: North has the highest total sales in this sample dataset at ■1,335,000. Compare regional totals with order counts and profit before making business decisions.

2. Monthly Sales Trend

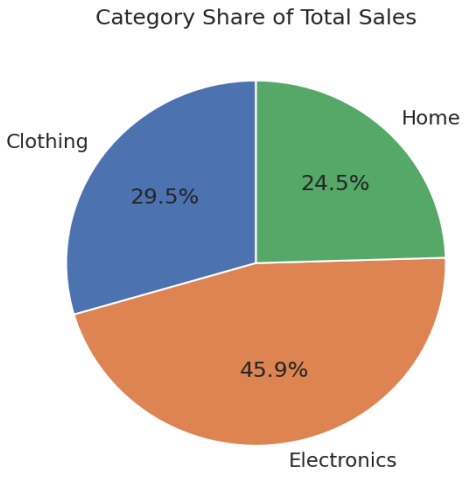


3. Sales by Product Category

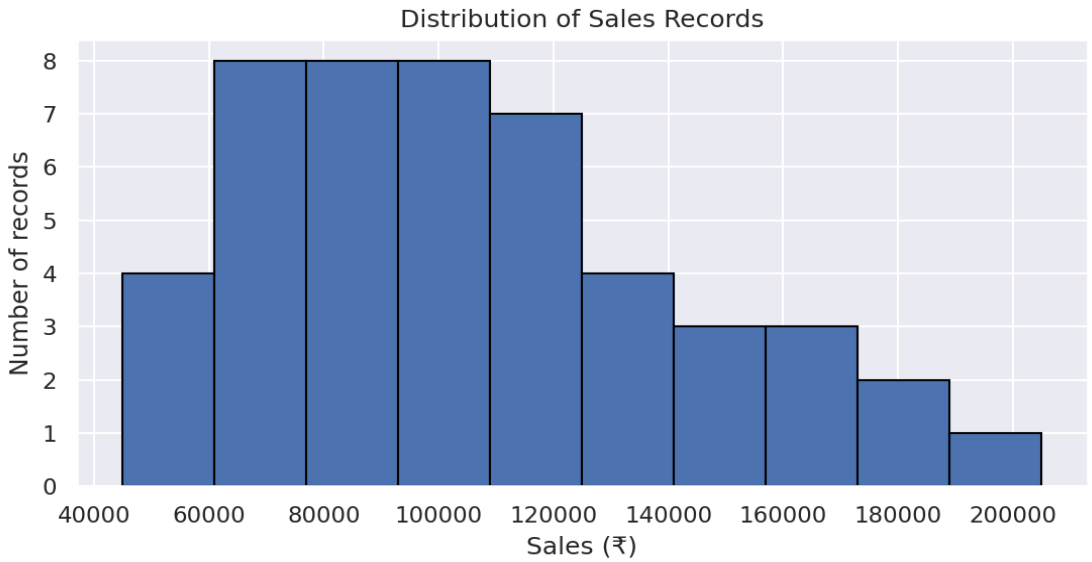


Finding: Electronics is the top category by total sales in this dataset.

4. Category Share



5. Distribution of Sales Records



The histogram shows how often different sales values occur. A distribution chart can help identify clusters and unusually high or low records.

6. Key Insights and Recommendations

• Total sales across 48 sample records: ■5,100,000.
• Highest sales region: North.
• Highest sales category: Electronics.
• The monthly line chart can help highlight changes across the year.
• Use category and regional comparisons to guide further questions, but do not assume causes without additional data.

Limitations

This is a synthetic dataset for learning, not actual company data. The charts demonstrate visualization techniques and should not be presented as real business performance.

How to run the project

Install dependencies with: python -m pip install pandas matplotlib
Then run: python create_dashboard.py
The script exports charts to the charts folder and summary CSV files to output.

Submission checklist: run the script, inspect the generated charts, upload the project files to your GitHub repository, and prepare a short LinkedIn video explaining your work with #codsoft.